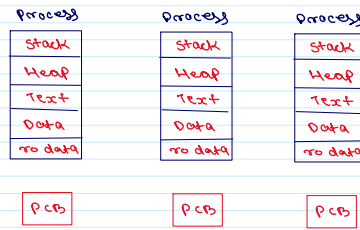
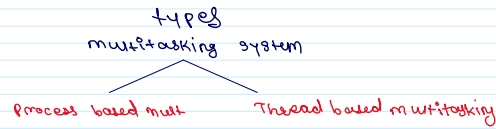
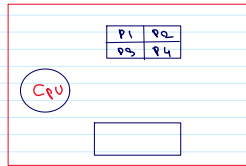
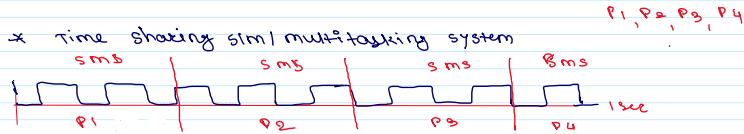


Day 2



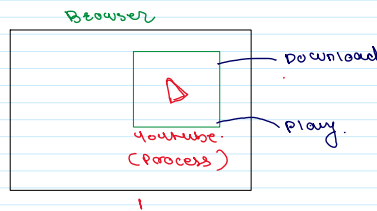
(process based multitasking)



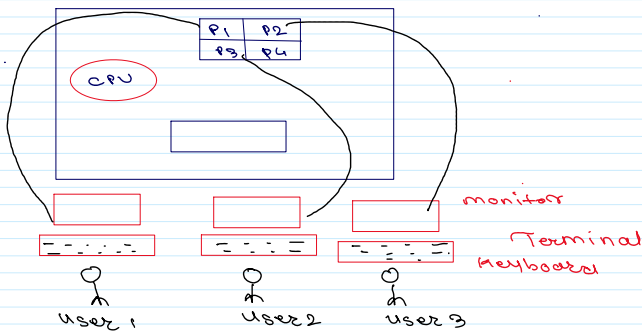
TCB - Thread Control block.
(Thread based multitasking)

Thread - Thread is a lightweight process
- it requires less resources
- Thread is a unit of execution.

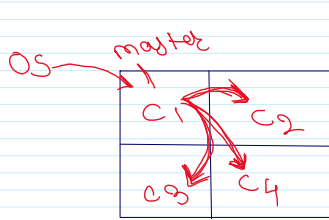
Process - is like a container for resources
- it requires more resources



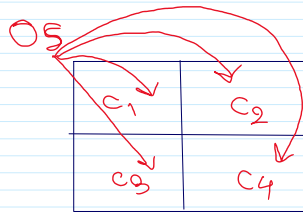
* multiuser system



⑥ multiprocessor system / multicore system.

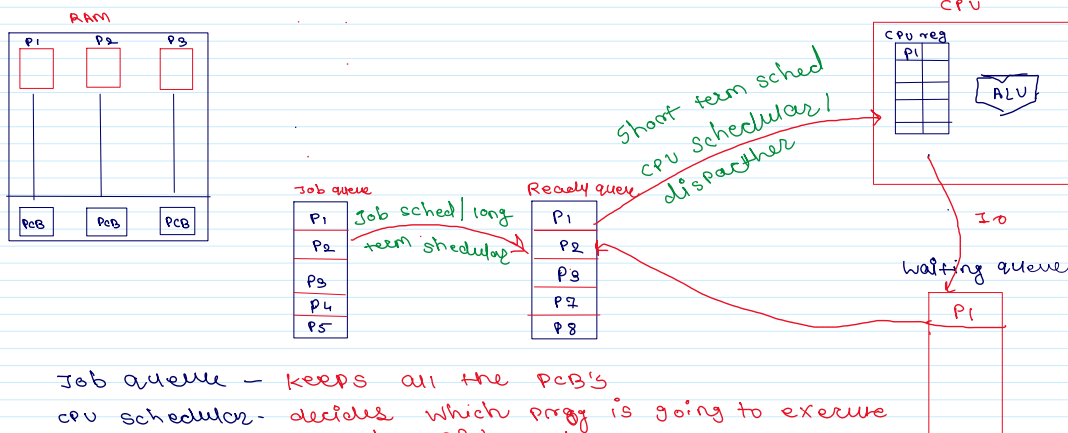


Asymmetric multicore sim.



Symmetric multicore sim.

* Process execution

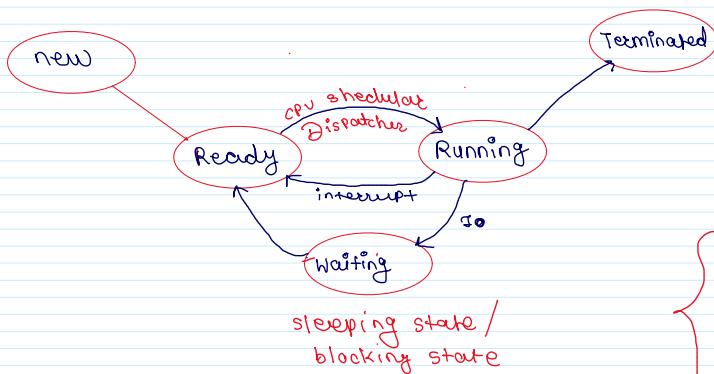


Job queue - keeps all the PCB's

cpu scheduler - decides which prog is going to execute on the CPU next

Dispatcher - loads prog ready queue to the CPU

* Process life cycle.

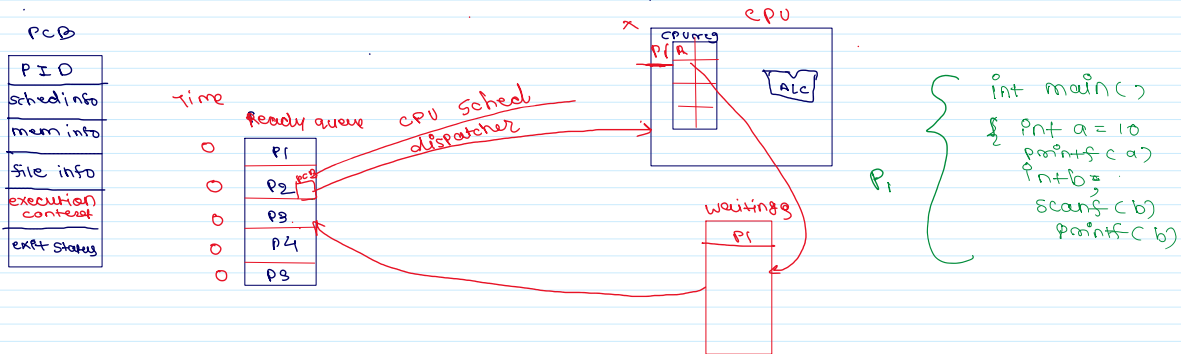


```
int main()
{
    while(1)
    {
        8 = 10 + 2;
    }
}
```

CPU Scheduler

- ① Running → Terminated } Non Preemptive
- ② Running → Waiting }
- ③ Running → Ready } Preemptive
- ④ Waiting → Ready }

* Context switching



Context switching - CPU is switching between one process PCB's execution context to the another process PCB's execution context